

Weinberg-operator coefficient

$$T3 - E, \quad \alpha = 0 : \quad S_1 = (3, 0), \quad S_2 = (3, 1), \quad F = (2, \tfrac{1}{2})$$

$$C_5 = \frac{2}{3} \left(- \frac{\sqrt{6} h M_F^3 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\sqrt{6} h M_F^3 M_{S1}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} - \frac{\sqrt{6} h M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S1}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} + \frac{\sqrt{6} h M_F M_{S1}^2 M_{S2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S2}^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S1}^2)(M_F^2 - M_{S2}^2)(M_{S1}^2 - M_{S2}^2)} \right)$$